Настройка сетевых сервисов

DNS и DHCP в Cisco Packet Tracer

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Цели и задачи работы

Цель лабораторной работы

Изучить принципы работы протоколов DNS и DHCP
и получить практические навыки их настройки
в среде Cisco Packet Tracer.

Настройка DNS-сервера

Добавление сервера в сеть

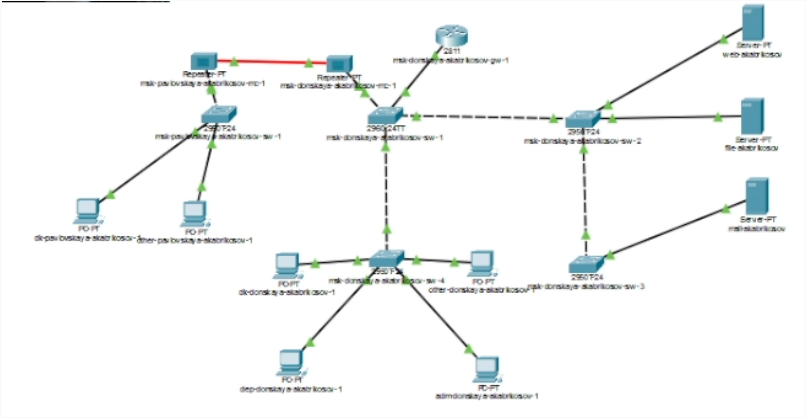


Рис. 1: Топология сети

Настройка IP-адреса сервера

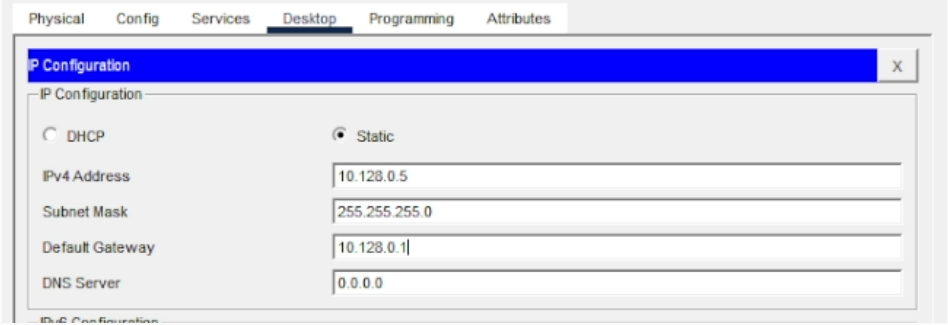


Рис. 2: IP-конфигурация DNS

Настройка DNS-записей

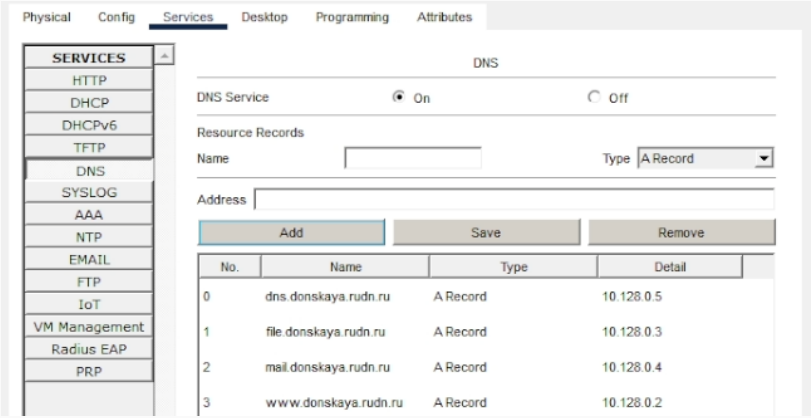


Рис. 3: DNS записи

Настройка DHCP

Конфигурация DHCP на маршрутизаторе

IOS Command Line Interface

```
Password:

msk-donskaya-akabrikosov-gw-1>enable
Password:
msk-donskaya-akabrikosov-gw-1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-akabrikosov-gw-1(config)#ip name-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(config)#service dhcp
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool dk
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.3.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#exit
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.3.1 10.128.3.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.3.200 10.128.3.254
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool departments
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.4.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.4.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#exit
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.4.1 10.128.4.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.4.200 10.128.4.254
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool dk
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#exit
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool adm
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.5.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.5.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#exit
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.5.1 10.128.5.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.5.200 10.128.5.254
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp pool other
msk-donskaya-akabrikosov-gw-1(dhcp-config)#network 10.128.6.0 255.255.255.0
msk-donskaya-akabrikosov-gw-1(dhcp-config)#default-router 10.128.6.1
msk-donskaya-akabrikosov-gw-1(dhcp-config)#dns-server 10.128.0.5
msk-donskaya-akabrikosov-gw-1(dhcp-config)#exit
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.6.1 10.128.6.29
msk-donskaya-akabrikosov-gw-1(config)#ip dhcp excluded-address 10.128.6.200 10.128.6.254
msk-donskaya-akabrikosov-gw-1(config)#
```

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Пулы DHCP

- dk — 10.128.3.0/24
- departments — 10.128.4.0/24
- adm — 10.128.5.0/24
- other — 10.128.6.0/24

Работа клиентов

Получение адреса по DHCP

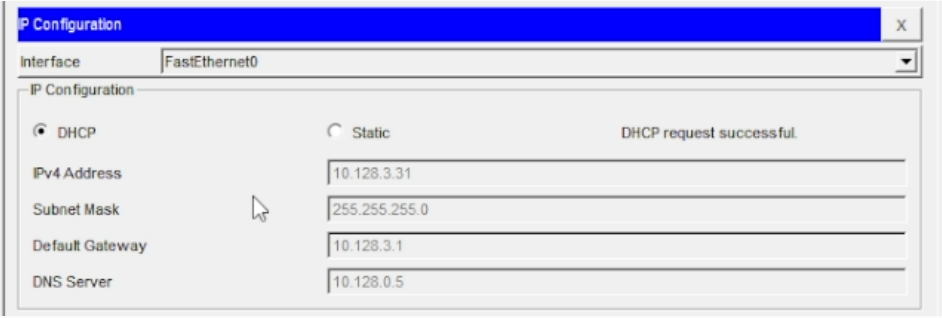


Рис. 5: DHCP на ПК

Информация о пулах

IOS Command Line Interface

```
 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.3.1           10.128.3.1      - 10.128.3.254      2    / 8     / 254

Pool departments :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.4.1           10.128.4.1      - 10.128.4.254      1    / 8     / 254

Pool adm :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.5.1           10.128.5.1      - 10.128.5.254      1    / 8     / 254

Pool other :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 8
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 10.128.6.1           10.128.6.1      - 10.128.6.254      2    / 8     / 254
msk-donskaya-akabrikosov-gw-1#
```

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Таблица выданных адресов

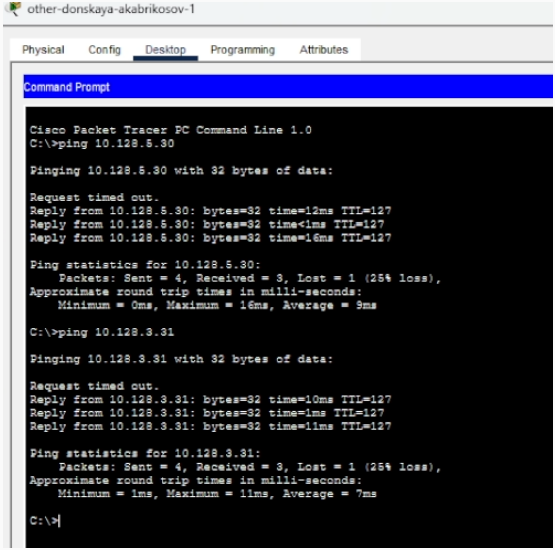
```
msk-donskaya-akabrikosov-gw-1#sh ip dhcp binding
IP address       Client-ID/                Lease expiration        Type
                 Hardware address
10.128.3.30      0005.5E99.13A3            --                      Automatic
10.128.3.31      0090.0C78.87B8            --                      Automatic
10.128.4.30      0006.2A60.E498            --                      Automatic
10.128.5.30      0007.EC22.6AD3            --                      Automatic
10.128.6.30      000A.F3AB.85C5            --                      Automatic
10.128.6.31      0060.2FB1.2DE9            --                      Automatic
msk-donskaya-akabrikosov-gw-1#
```

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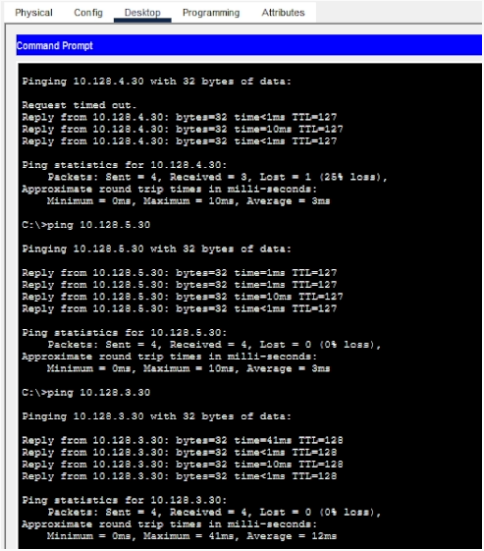
Рис. 7: DHCP binding

Проверка сети

Проверка доступности



Связь между подсетями



Работа DHCP (Simulation)

Процесс DHCP

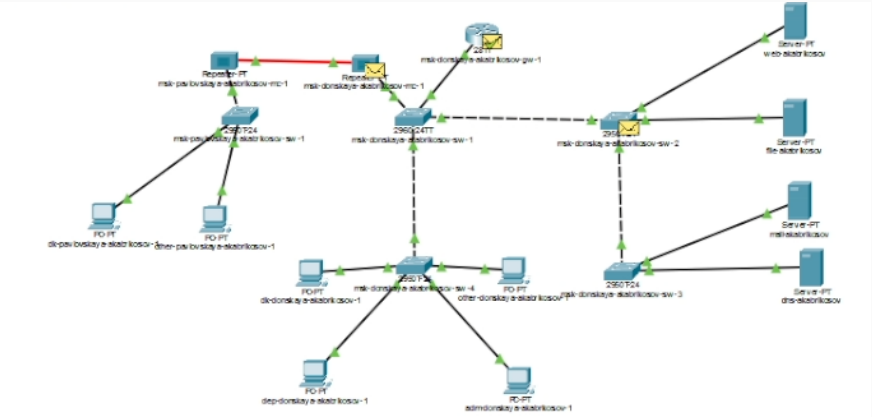


Рис. 10: DHCP Simulation

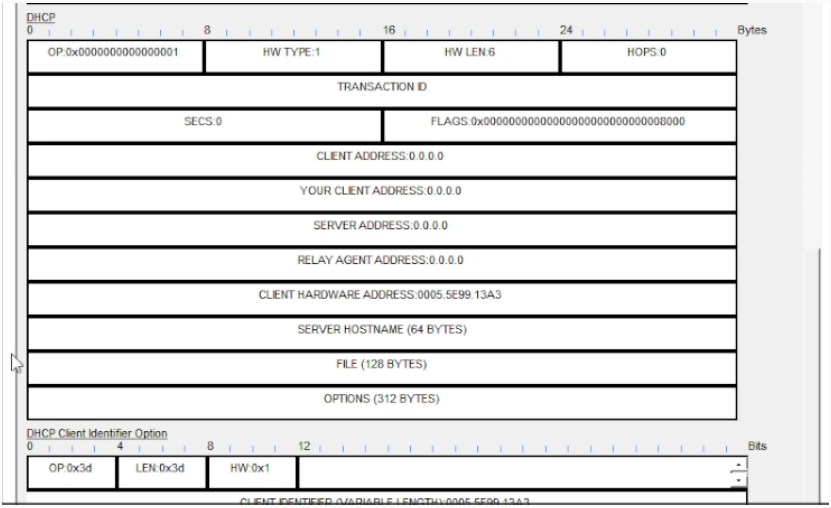
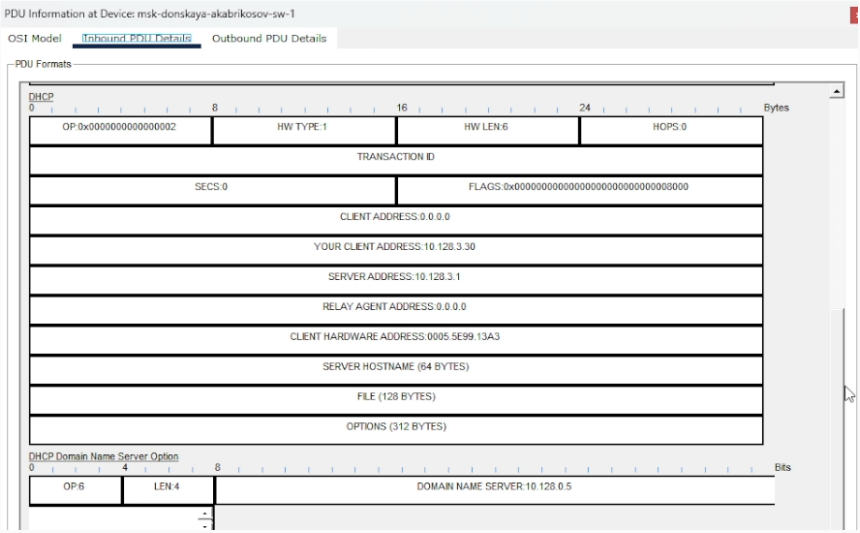


Рис. 11: DHCP Discover

Offer


PDU Information at Device: msk-donskaya-akabrikosov-sw-1
OSI Model
Inbound PDU Details
Outbound PDU Details
PDU Formats
DHCP
0
8
16
24
Bytes
OP:0x0000000000000002
HW TYPE:1
HW LEN:6
HOPS:0
TRANSACTION ID
SECS:0
FLAGS:0x00000000000000000000000000008000
CLIENT ADDRESS:0.0.0.0
YOUR CLIENT ADDRESS:10.128.3.30
SERVER ADDRESS:10.128.3.1
RELAY AGENT ADDRESS:0.0.0.0
CLIENT HARDWARE ADDRESS:0005.5E99.13A3
SERVER HOSTNAME (64 BYTES)
FILE (128 BYTES)
OPTIONS (312 BYTES)
DHCP Domain Name Server Option
0
4
8
Bits
OP:6
LEN:4
DOMAIN NAME SERVER:10.128.0.5

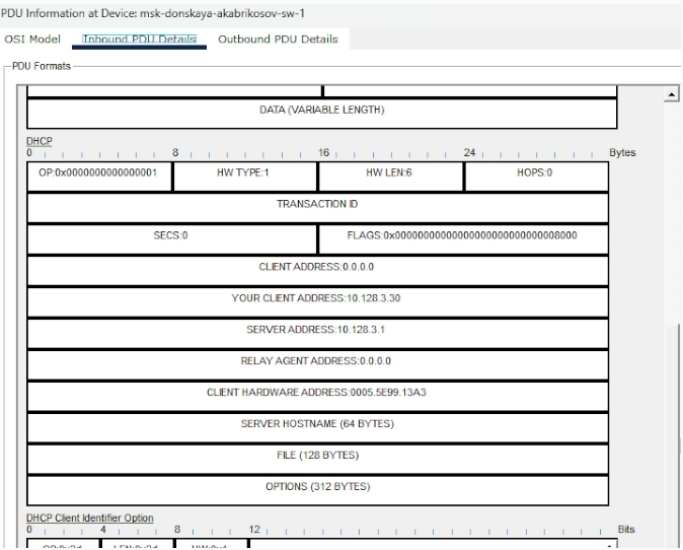


Рис. 13: DHCP Request

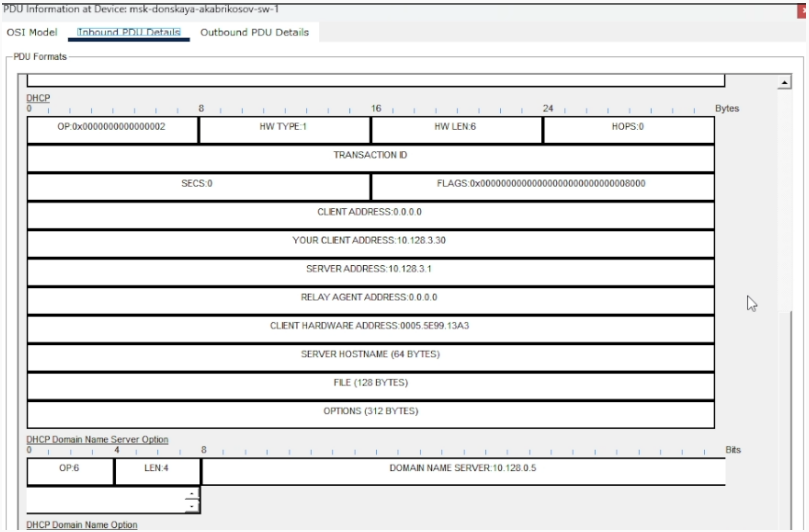


Рис. 14: DHCP ACK

Вывод

Итоги работы

Настроены сервисы DNS и DHCP в сети Cisco Packet Tracer.

Реализовано автоматическое распределение IP-адресов
и разрешение доменных имён.

Исследован процесс DHCP и структура сообщений.

Подтверждена работоспособность сети и взаимодействие подсетей.